

Survey report on emerging trends in low-code application development.

1. AIM

To study the emerging trends in low-code application development and analyze how low-code platforms can be used to develop a Gym Membership Enrolment and Front Desk Management application.

The application is designed using Zoho Creator to replace the traditional paper-based gym registration process with an online digital form. It captures member information such as name, email, phone number, address, emergency contact, age, membership plan, preferred activities, fitness goals, preferred start date, workout time, and health declaration.

The study also aims to compare traditional application development with low-code development and examine the benefits, challenges, and features of existing low-code platforms.

2. TRADITIONAL APPLICATION DEVELOPMENT VS EMERGING TRENDS IN LOW-CODE APPLICATION DEVELOPMENT

Traditional Application Development

In traditional application development, applications are developed by writing a large amount of source code using programming languages and frameworks. Developers are responsible for designing the user interface, implementing business logic, connecting databases, testing, debugging, and deployment.

For example, developing a gym membership system traditionally would require developers to create:

- Front-end web pages
- Form validation

- Database tables
- Backend APIs
- Authentication
- Report generation
- Email notifications
- Deployment infrastructure

This approach provides extensive customization but generally requires more development time and programming knowledge.

Low-Code Application Development

Low-code application development uses visual development environments, drag-and-drop components, pre-built forms, workflows, database integrations, and automation tools to reduce the amount of manually written code.

For the given application, Zoho Creator provides a Form Builder containing Basic Fields. The assignment specifically requires the Gym Membership Enrolment Form to use all 13 Basic Field types.

The application contains fields such as:

- Name
- Email
- Address
- Phone
- Single Line
- Multi Line
- Number
- Date
- Time
- Drop Down
- Radio
- Multi Select

- Checkbox

Emerging Trends in Low-Code Development

The major emerging trends include:

1. Visual Application Builders – Applications can be constructed using drag-and-drop components.
2. Workflow Automation – Repetitive processes can be automated.
3. AI-Assisted Development – AI is increasingly being used to generate application components and assist developers.
4. Cloud-Based Development – Applications can be developed and accessed through cloud platforms.
5. API and Third-Party Integration – Low-code platforms increasingly provide connectors for external services.
6. Mobile Application Development – Applications can be created for both web and mobile environments.
7. Rapid Prototyping – Functional prototypes can be developed quickly.
8. Citizen Development – Non-professional programmers can participate in application development.
9. Data-Driven Applications – Forms, databases, reports, and dashboards can be integrated within one platform.
10. Automation and Notifications – Events such as form submissions can trigger automated actions.

For the gym application, the workflow demonstrates this approach: the member submits the form, Zoho Creator validates the fields, creates a report record, sends a confirmation email, and allows the front desk to filter and manage enrolment records.

3. BENEFITS AND CHALLENGES IN LOW-CODE

Benefits of Low-Code Development

1. Faster Development

Low-code platforms reduce development time because developers can use pre-built components rather than creating every component from scratch.

2. Less Coding

Basic application functionality can be implemented using configuration and visual tools.

3. Easy Form Development

Forms can be created using predefined fields such as text, email, phone, number, date, dropdown, radio button, multi-select, and checkbox.

4. Rapid Prototyping

A working prototype can be created quickly and modified based on user requirements.

5. Workflow Automation

Actions such as email notifications and record creation can be automated.

6. Easy Data Management

Form submissions can automatically be stored and displayed through reports.

7. Reduced Development Complexity

Developers do not need to manually implement every basic component of the application.

8. Accessibility

Low-code platforms allow both developers and technically skilled business users to participate in application development.

Application Example

In the Gym Membership application, once a member submits the form, the platform validates mandatory fields and formats, creates a new record in the membership report, and triggers a confirmation email.

Challenges of Low-Code Development

1. Platform Dependency

Applications may become dependent on the selected low-code platform.

2. Limited Customization

Highly specialized requirements may be difficult to implement compared with completely custom software.

3. Security Concerns

Sensitive information must be protected through proper authentication, authorization, and security configuration.

4. Scalability

Some applications may require careful planning when the number of users and data increases significantly.

5. Vendor Lock-In

Moving an application from one low-code platform to another may require significant redevelopment.

6. Limited Control

Developers may have less control over the underlying infrastructure compared with traditional development.

7. Learning Platform-Specific Tools

Although programming requirements are reduced, users still need to learn the platform's interface, configuration, and workflow system.

8. Integration Limitations

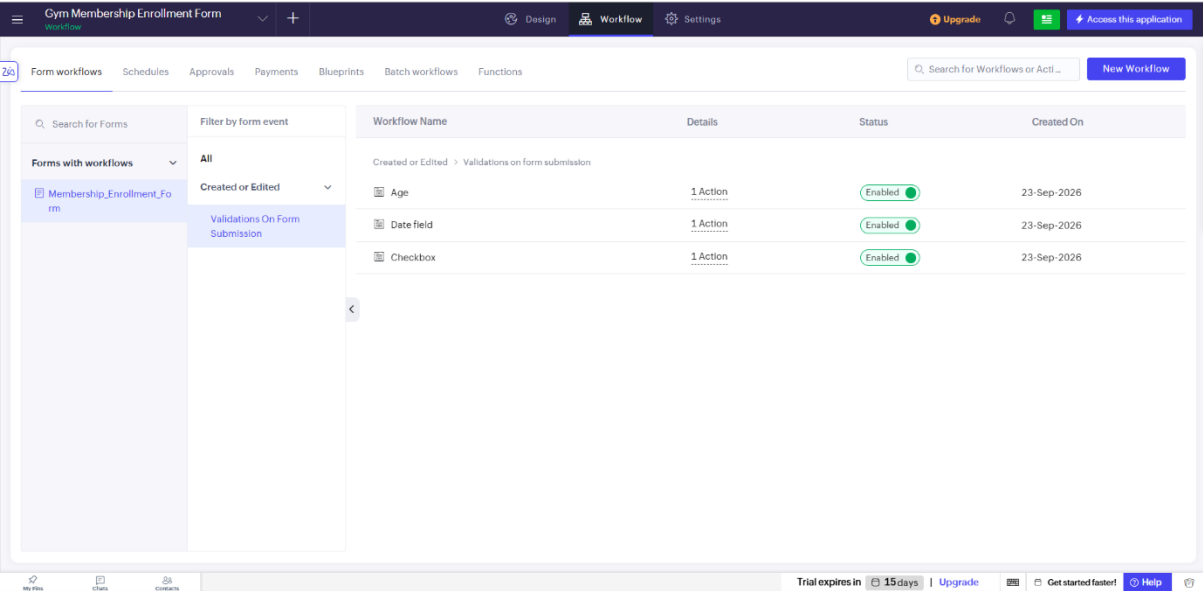
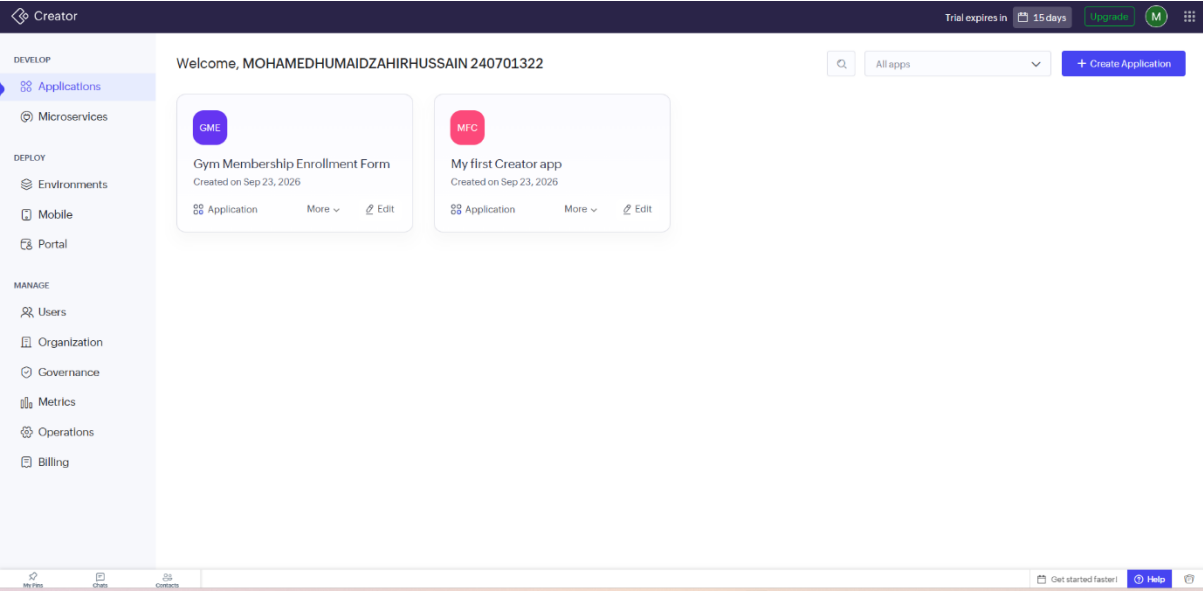
Complex integrations with external systems may require APIs or custom code.

4. COMPARISON OF EXISTING LOW-CODE PLATFORMS

Feature	Zoho Creator	Microsoft Power Apps	Mendix	OutSystems	Google AppSheet
Development approach	Low-code visual development	Low-code application development	Enterprise low-code	Enterprise low-code	No-code/low-code
Form creation	Yes	Yes	Yes	Yes	Yes
Workflow automation	Yes	Yes	Yes	Yes	Yes
Database support	Built-in/application data	Dataverse and connectors	Integrated data modeling	Integrated data services	Spreadsheet/database-oriented
Reporting	Reports and dashboard	Reports/dashboards through ecosystem	Dashboards	Dashboards	Reports/views
Mobile applications	Supported	Supported	Supported	Supported	Supported
Third-party integration	Supported	Strong integration ecosystem	Supported	Supported	Supported
AI capabilities	Platform-dependent	Integrated Microsoft AI ecosystem	AI-assisted capabilities	AI-assisted development	AI-assisted features
Suitable for	Business applications and internal systems	Business applications and enterprise workflows	Enterprise applications	Enterprise-scale applications	Data-driven business apps

Coding requirement	Low	Low	Low	Low	Very low for basic applications
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Screenshots



Member_Name	Email_Address	Residential_Address	Contact_Number	Emergency_Contact_Name	Fitness_Goals_Medical_Notes	Age
Vignesh Raja	vignesh.raja@gmail.com	14, Bharathi Nagar, Chennai, Tamil Nadu, 638001, India	+919047088990	Priya Raja	Aiming for weight loss and improved stamina; has mild knee discomfort from a past injury.	31

Gym Membership Enrollment Form

- Membership_Enrollment_Form
- Membership_Enrollment_Form
- Membership_Enrollment_Form Report

Membership_Enrollment_Form

Member Details

Member_Name

First Name Last Name

Age

Membership_Type

☐ New Member

☐ Renewal

☐ Corporate Enrollment

Contact Details

Email_Address

Contact_Number

Residential_Address

Address Line 1

Address Line 2

City / District

State / Province

Postal Code

Country

Emergency_Contact_Name

Membership & Fitness Preferences

Membership_Plan

Preferred_Activities

Fitness_Goals_Medical_Notes

Scheduling & Health Declaration

Preferred_Start_Date

Preferred_Workout_Slot

Health_Declaration_Waiver ☐ I declare my fitness to exercise and accept the liability waiver

Submit

5. RESULT

The survey on low-code application development was successfully completed. The Gym Membership Enrolment application demonstrates how Zoho Creator can simplify form creation, validation, data management, and workflow automation.